

Sample Collection: students

Use the following sample data for all tasks.

```
db.students.insertMany([
```

```
{student_id:1,
```

```
name:"Saran",
```

```
department:"CSE",
```

```
city:"Bangalore",
```

```
cgpa:8.5,
```

```
age:21,
```

```
gender:"Male"},
```

```
{student_id:2,
```

```
name:"Priya",
```

```
department:"ECE",
```

```
city:"Hyderabad",
```

```
cgpa:9.0,
```

```
age:22,
```

```
gender:"Female"},
```

```
{student_id:3,
```

```
name:"Rahul",
```

```
department:"CSE",
```

```
city:"Chennai",
```

```
cgpa:7.8,
```

```
age:21,
```

```
gender:"Male"},
```

```
{student_id:4,
```

```
name:"Deepa",
```

```
department:"EEE",
```

```
city:"Bangalore",
```

```
cgpa:8.9,
```

age:22,
gender:"Female"},
{student_id:5,
name:"Anjali",
department:"ECE",
city:"Pune",
cgpa:8.2,
age:20,
gender:"Female"},
{student_id:6,
name:"Kumar",
department:"Mechanical",
city:"Hyderabad",
cgpa:7.5,
age:23,
gender:"Male"},
{student_id:7,
name:"Sneha",
department:"CSE",
city:"Bangalore",
cgpa:9.1,
age:21,
gender:"Female"},
{student_id:8,
name:"Varun",
department:"Civil",
city:"Chennai",
cgpa:8.0,
age:22,

```
gender:"Male"},
{student_id:9,
name:"Charan",
department:"Mechanical",
city:"Pune",
cgpa:7.2,
age:23,
gender:"Male"},
{student_id:10,
name:"Harsha",
department:"CSE",
city:"Hyderabad",
cgpa:8.7,
age:21,
gender:"Male"}]);
```

Commands:

```
use MITS
switched to db MITS
use MITS
already on db MITS
db.students.insertMany([
{student_id:1,
name:"Saran",
department:"CSE",
city:"Bangalore",
cgpa:8.5,
age:21,
gender:"Male"},
{student_id:2,
```

```
name:"Priya",
department:"ECE",
city:"Hyderabad",
cgpa:9.0,
age:22,
gender:"Female"},
{student_id:3,
name:"Rahul",
department:"CSE",
city:"Chennai",
cgpa:7.8,
age:21,
gender:"Male"},
{student_id:4,
name:"Deepa",
department:"EEE",
city:"Bangalore",
cgpa:8.9,
age:22,
gender:"Female"},
{student_id:5,
name:"Anjali",
department:"ECE",
city:"Pune",
cgpa:8.2,
age:20,
gender:"Female"},
{student_id:6,
name:"Kumar",
```

department:"Mechanical",
city:"Hyderabad",
cgpa:7.5,
age:23,
gender:"Male"},
{student_id:7,
name:"Sneha",
department:"CSE",
city:"Bangalore",
cgpa:9.1,
age:21,
gender:"Female"},
{student_id:8,
name:"Varun",
department:"Civil",
city:"Chennai",
cgpa:8.0,
age:22,
gender:"Male"},
{student_id:9,
name:"Charan",
department:"Mechanical",
city:"Pune",
cgpa:7.2,
age:23,
gender:"Male"},
{student_id:10,
name:"Harsha",
department:"CSE",

```
city:"Hyderabad",
cgpa:8.7,
age:21,
gender:"Male"}}])
```

COUNT Aggregation Tasks

Task 1

Find the total number of students.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    totalStudents: { $sum: 1 } } }])
```

Task 2

Find the number of students in each department.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    count: { $sum: 1 } } }])
```

Task 3

Find the number of students in each city.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    count: { $sum: 1 } } }])
```

Task 4

Find the number of male students.

Commands:

```
db.students.aggregate([
  { $match: { gender: "Male" } },
  { $group: {
    _id: null,
    count: { $sum: 1 } } }])
```

Task 5

Find the number of female students.

Commands:

```
db.students.aggregate([
  { $match: { gender: "Female" } },
  { $group: {
    _id: null,
    count: { $sum: 1 } } }])
```

AVG Aggregation Tasks

Task 6

Find the average CGPA of all students.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    averageCGPA: { $avg: "$cgpa" } } }])
```

Task 7

Find the average CGPA department-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    averageCGPA: { $avg: "$cgpa" } } }])
```

Task 8

Find the average age department-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    averageAge: { $avg: "$age" } } })
```

Task 9

Find the average CGPA city-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    averageCGPA: { $avg: "$cgpa" } } })
```

Task 10

Find the average age of all students.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    averageAge: { $avg: "$age" } } })
```

SUM Aggregation Tasks

Task 11

Find the sum of all CGPAs.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    totalCGPA: { $sum: "$cgpa" } } })
```

Task 12

Find the total age of all students.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    totalAge: { $sum: "$age" } } }])
```

Task 13

Find the total CGPA department-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    totalCGPA: { $sum: "$cgpa" } } }])
```

Task 14

Find the total age city-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    totalAge: { $sum: "$age" } } }])
```

Task 15

Find the total CGPA of CSE students.

Commands:

```
db.students.aggregate([
  { $match: { department: "CSE" } },
  { $group: {
    _id: null,
    totalCGPA: { $sum: "$cgpa" } } }])
```

MAX Aggregation Tasks

Task 16

Find the highest CGPA.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    highestCGPA: { $max: "$cgpa" } } })
```

Task 17

Find the highest age.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    highestAge: { $max: "$age" } } })
```

Task 18

Find the highest CGPA department-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    highestCGPA: { $max: "$cgpa" } } })
```

Task 19

Find the highest age city-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    highestAge: { $max: "$age" } } })
```

Task 20

Find the department having the maximum average CGPA.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    averageCGPA: { $avg: "$cgpa" } } },
  { $sort: { averageCGPA: -1 } },
  { $limit: 1 } ]])
```

MIN Aggregation Tasks

Task 21

Find the minimum CGPA.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    minimumCGPA: { $min: "$cgpa" } } } ]])
```

Task 22

Find the minimum age.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: null,
    minimumAge: { $min: "$age" } } } ]])
```

Task 23

Find the minimum CGPA department-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
```

```
minimumCGPA: { $min: "$cgpa" } } }])
```

Task 24

Find the minimum age city-wise.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    minimumAge: { $min: "$age" } } } ]])
```

Task 25

Find the city having the lowest average CGPA.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    averageCGPA: { $avg: "$cgpa" } } },
  { $sort: { averageCGPA: 1 } },
  { $limit: 1 } ]])
```

GROUP BY Equivalent (\$group)

Task 26

Group students by department and count them.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    count: { $sum: 1 } } } ]])
```

Task 27

Group students by city and count them.

Commands:

```
db.students.aggregate([
```

```
{ $group: {
  _id: "$city",
  count: { $sum: 1 } } }])
```

Task 28

Group students by gender and count them.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$gender",
    count: { $sum: 1 } } } ])
```

Task 29

Group students by department and find average CGPA.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    averageCGPA: { $avg: "$cgpa" } } }])
```

Task 30

Group students by city and find average age.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    averageAge: { $avg: "$age" } } }])
```

MATCH + GROUP

Task 31

Find the number of students with CGPA greater than 8.

Commands:

```
db.students.aggregate([
```

```
{ $match: { cgpa: { $gt: 8 } } },
{ $group: {
  _id: null,
  count: { $sum: 1 } } } ]])
```

Task 32

Find the average CGPA of CSE students.

Commands:

```
db.students.aggregate([
  { $match: { department: "CSE" } },
  { $group: {
    _id: null,
    averageCGPA: { $avg: "$cgpa" } } } ]])
```

Task 33

Find the total number of female students.

Commands:

```
db.students.aggregate([
  { $match: { gender: "Female" } },
  { $group: {
    _id: null,
    totalFemales: { $sum: 1 } } } ]])
```

Task 34

Find the highest CGPA among male students.

Commands:

```
db.students.aggregate([
  { $match: { gender: "Male" } },
  { $group: {
    _id: null,
    highestCGPA: { $max: "$cgpa" } } } ]])
```

Task 35

Find the lowest CGPA among female students.

Commands:

```
db.students.aggregate([
  { $match: { gender: "Female" } },
  { $group: {
    _id: null,
    lowestCGPA: { $min: "$cgpa" } } })
```

SORT Aggregation

Task 36

Display departments ordered by average CGPA in descending order.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    averageCGPA: { $avg: "$cgpa" } } },
  { $sort: { averageCGPA: -1 } })
```

Task 37

Display cities ordered by student count.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    count: { $sum: 1 } } },
  { $sort: { count: -1 } })
```

Task 38

Display departments ordered by maximum CGPA.

Commands:

```
db.students.aggregate([
  { $group: {
```

```
  _id: "$department",
  maxCGPA: { $max: "$cgpa" }   } },
{ $sort: { maxCGPA: -1 } }])
```

Task 39

Display cities ordered by average age.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$city",
    averageAge: { $avg: "$age" }   } },
  { $sort: { averageAge: -1 } }])
```

Task 40

Display departments ordered by number of students.

Commands:

```
db.students.aggregate([
  { $group: {
    _id: "$department",
    count: { $sum: 1 }   } },
  { $sort: { count: -1 } }])
```